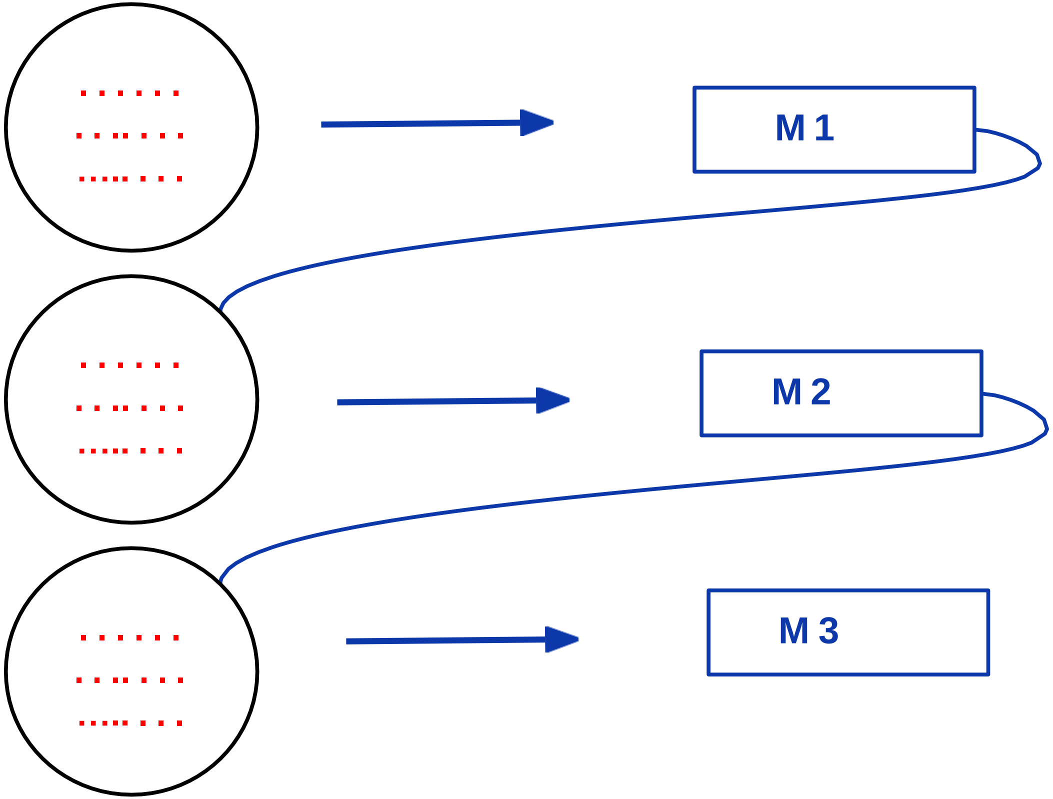
Boosting



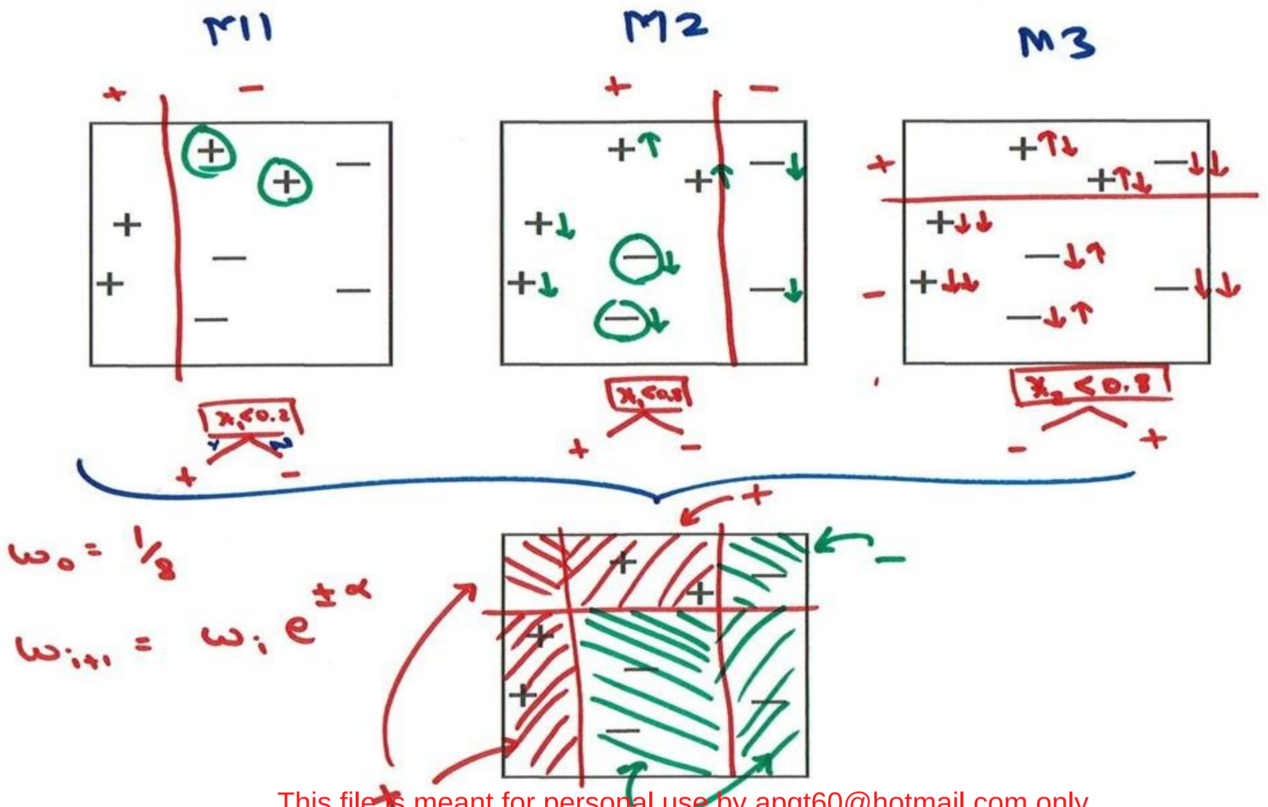
Boosting Methods

- **AdaBoosting** (Adaptive Boosting)
  - In AdaBoost, the successive learners are created with a focus on the ill fitted data of the previous learner
  - Each successive learner focuses more and more on the harder to fit data i.e. their residuals in the previous tree
- **Gradient Boosting** (GBM)
  - Each learner is fit on a modified version of original data. Original data is replaced with the x values and residuals from previous learner
  - By fitting new models to the residuals, the overall learner gradually improves in areas where residuals are initially high
- **XG Boost (Extreme Gradient Boosting)** (XGM)
  - Upgraded implementation of Gradient Boosting. Developed for high computational speed, scalability, and better performance.
  - Parallel Implementation, Cross-Validation, Cache Optimization, Distributed Computation

AdaBoost

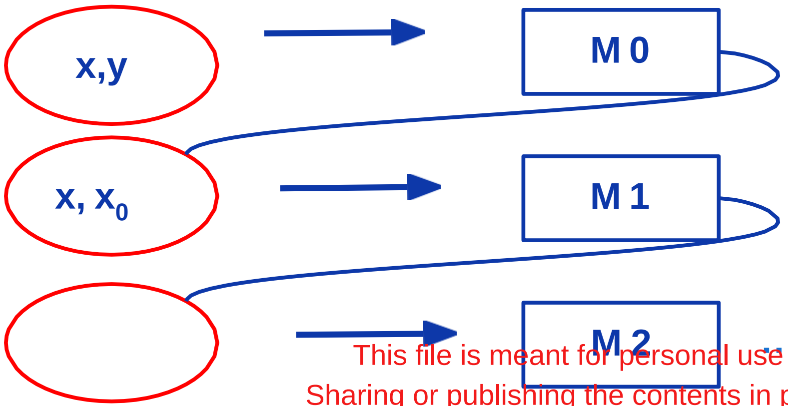
X1	X2	Y
...	...	+
..	...	+
..	...	-
..	...	...
..	...	....
..	...	....
..	...	....
..	...	.....
..	...	....
..	...	....
..	...	....
..	...	.....

AdaBoost



Gradient Boosting

X			y	y_0	$y - y_0$	h
			50	40	10	8
			92	100	-8	-8
			60	80	-20	-10
			64	50	14	12
			....	....	.....	......



$$y_0 = h_0 (x)$$

$$y = h_0 (x) + \alpha_1 h_1 (x)$$

$$y = h_0 (x) + \alpha_1 h_1 (x) + \alpha_2 h_2 (x) + ...$$

Stacking

Parallelly, Independent

Weak Learners

Bagging

Boosting

Stacking

Homogeneous Model

Combine using voting or avg

Seq, adaptively

Computationally Efficient

Heterogeneous Model

Combine using another meta model

Train Test

Fold 1 | Fold 2 |

KNN → Pred on Fold 2

Log. Reg → Pred on Fold 2 → NN → Predict

SVM → Pred on Fold 2

Any model can be used to create a stacking model not only KNN, SVM, Logistic regression, or NN